

# Markov Random Fields with Spacial Seeds

## Bayesian Estimation on Hierarchical Hidden Field Model

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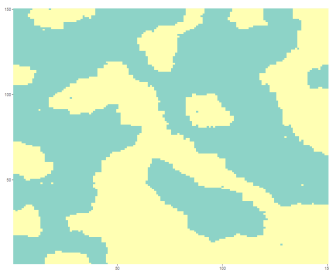
University of Campinas

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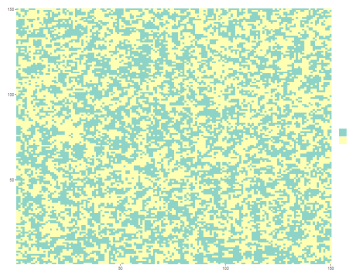
# 1. Introduction

# The Potts Model

- Spatial models on regular lattices (pixels) often employ **Markov Random Fields (MRF)**.
- The **Potts Model** is the classical standard for capturing local neighborhood dependence.



(a) High neighborhood dependence



(b) Low neighborhood dependence

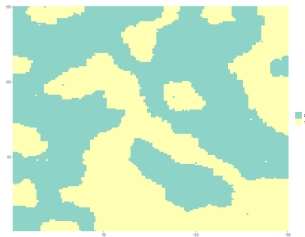
# The Potts Model

- **Energy Function:** For the traditional Potts Model, we define a spatial field  $Z$  where the local probability of pixel  $i$  taking class  $k$  is proportional to  $\exp(\eta_i(k))$ :

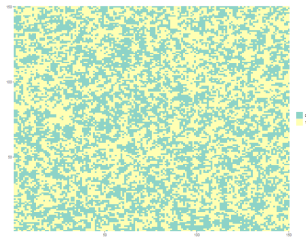
$$\eta_i(k) = \alpha \sum_{j \in N(i)} \mathbb{1}(Z_j = k) + \beta \mathbb{1}(k = 0)$$

- **Parameters Meaning:**
  - $N(i)$ : Pixel  $i$  neighbors.
  - $\alpha$  (**Uniformity**): Local attraction between neighboring pixels (smoothing).
  - $\beta$  (**Global Abundance**): Force favoring/penalizing the background (class 0).

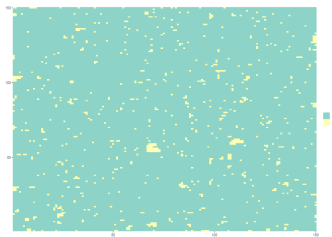
# The Potts Model



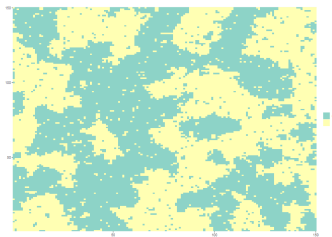
(a)  $\alpha = 2, \beta = 0$



(b)  $\alpha = 0.5, \beta = 0$



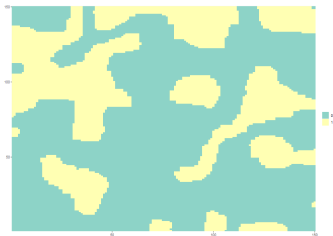
(c)  $\alpha = 1, \beta = 0.1$



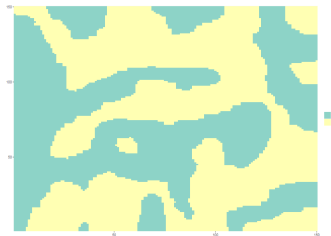
(d)  $\alpha = 1, \beta = 0$

# The Spacial Limitation

- Many real images and textures possess spacial patterns we call “seeds” that may attract or repel colors.
- **Our Proposal:** Incorporate spacial seeds that exert radial influence on the classes.
- **Objective:** Propose a reparameterized model that describe the inclusion of centroids/seeds, and validate its inference under a hidden model.



(a)  $\alpha = 3, \beta = 0$



(b)  $\alpha = 3, \beta = 0$

## 2. The Proposal

# Potts Reparameterized with Seeds

- **Enhanced Local Energy Function:** To address the isotropic limitation of the Potts model, we introduce a few elements to handle the spacial centrality bias in our local energy function:

$$\eta_i(k) = \alpha \sum_{j \in N(i)} \mathbb{1}(Z_j = k) + \beta \mathbb{1}(k = 0) + \sum_s \frac{\delta_s}{1 + d(i, s)} \mathbb{1}(c_s = k)$$

- **Parameters Meaning:**
  - $\alpha$  (**Uniformity**): Local attraction between neighboring pixels (smoothing).
  - $\beta$  (**Global Abundance**): Force favoring/penalizing the background (class 0).
  - $\delta_s$  (**Centrality**): Radial attraction/repulsion of the seed  $s$ .
  - $d(i, s)$ : Distance between pixel  $i$  and seed  $s$ .
  - $c_s$ : Class/color related to the seed.



# Potts Reparameterized with Seeds

- Enhanced Local Energy Function:**

$$\eta_i(k) = \alpha \sum_{j \in N(i)} \mathbb{1}(Z_j = k) + \beta \mathbb{1}(k = 0) + \sum_s \frac{\delta_s}{1 + d(i, s)} \mathbb{1}(c_s = k)$$

- Enhanced Global Energy Function:**

$$U(\omega) = \alpha \sum_{\{i,j\} \in E} \sum_{k \in S} \mathbb{1}(\omega_i = k) \mathbb{1}(\omega_j = k) + \beta \sum_{i \in N} \mathbb{1}(\omega_i = 0) + \sum_{i \in N} \sum_s \frac{\delta_s}{1 + d(i, s)} \mathbb{1}(\omega_i = c_s)$$

# The Threshold of the “Area of Influence”

- **Equilibrium of Forces:** The local seed attraction competes against the global abundance of the background class.
- **Null Boundary:** Under our model, ignoring neighborhood smoothing, they cancel out when:

$$\eta_i(0) = \eta_i(k) \implies d_{i,s} = \frac{\delta_s}{\beta} - 1$$

- **Spacial Transition:** That “Null Boundary” radius gives to the model a rough outline of the objects in the image.
  - **Within radius  $d_i$ :** The seed attraction dominates the local probability.
  - **Beyond radius  $d_i$ :** The class 0 dominates.
  - **Smoothing effect:** Depending on  $\alpha$  value, pixels can be affected more by its neighbors than by the radius rule.

# The Theoretical Bottleneck: Intractable Likelihood

- **The Gibbs Measure:** The global probability of an image configuration  $\omega$  is governed by:

$$P(\omega) = \frac{1}{Z(\theta)} \exp\{U(\omega)\}$$

- **The Computational Bottleneck:** The term  $Z(\theta)$  is the **normalizing constant** (partition function) summed over all lattice configurations  $\Omega$ :

$$Z(\theta) = \sum_{\omega \in \Omega} \exp\{U(\omega)\}$$

- For a small  $50 \times 50$  grid and  $K = 3$  colors, this requires  $3^{2500}$  terms, rendering standard Maximum Likelihood impossible.

## 3. Hidden & Hierarchical Models

# The Major Evolution: The Hidden Model (HMRF)

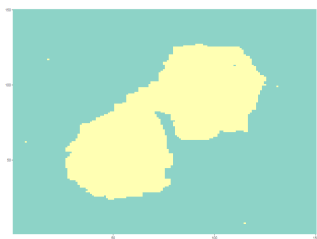
- **The Latent Field:** In segmentation practice, the label field  $Z$  (the true texture) is **latent/hidden**.
- **Observed Intensities:** We only observe the noisy, continuous intensities  $Y$ .
- **Observation Equation (Emission):** Conditionally on the latent label  $Z_i = k$ , the intensity  $Y_i$  is Normal:

$$Y_i \mid Z_i = k \sim N(\mu_k, \sigma^2)$$

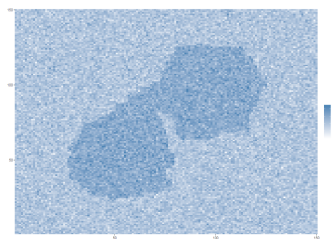
- **Normal Mixture:** Marginalizing over  $Z_i$ ,  $Y_i$  forms a spatial mixture of Gaussians:

$$f(y_i) = \sum_{k=0}^{K-1} P(Z_i = k) \phi(y_i; \mu_k, \sigma^2)$$

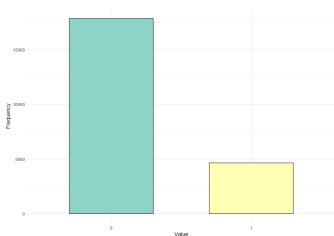
# The Major Evolution: The Hidden Model (HMRF)



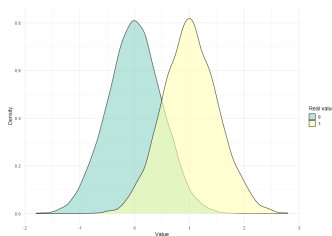
(a) 2-seeds generated image



(b) Same image plus  $N(0, 0.5)$  noise



(c) Clean image color frequencies



(d) Noisy image color densities

# Hierarchical Bayesian Formulation

- **Full Joint Distribution:** Treating  $Z$ , parameters  $\theta$ , and seed coordinates  $S$  as random variables:

$$P(Y, Z, \theta) = P(Y \mid Z, \mu, \sigma^2) P(Z \mid \alpha, \beta, \delta, S) \times \\ \pi(\alpha) \pi(\beta) \prod_{j=1}^M \pi(s_j) \pi(\delta_j) \prod_{k=0}^{K-1} \pi(\mu_k) \pi(\sigma^2)$$

- **Prior Distributions:**

- **Field (Gamma):**  $\alpha \sim \text{Gamma}(a_\alpha, b_\alpha)$ ,  $\beta \sim \text{Gamma}(a_\beta, b_\beta)$ .
- **Means (Normal):**  $\mu_k \sim N(m_k, \tau_k^2)$  (anchored via K-means).
- **Variances (Inverse-Gamma):**  $\sigma^2 \sim \text{IG}(a_\sigma, b_\sigma)$ .
- **Spatial:** Seed position  $s_j \sim \text{Unif}(L)$  discretized on the grid.

## 4. Estimation & Algorithms



# The Metropolis-Hastings Step via Pseudoposterior

- **Fixing the Bottleneck:** Because the true likelihood is intractable, we replace  $P(Z \mid \theta)$  with the Pseudolikelihood  $PL(\theta; Z)$  in the Metropolis-Hastings acceptance step.

$$P(Z_i = k \mid Z_{-i}, \theta) = \frac{P(Z_i = k, Z_{-i} \mid \theta)}{P(Z_{-i} \mid \theta)} = \frac{\frac{1}{Z(\theta)} \exp\{U(Z_i = k, Z_{-i})\}}{\sum_{\ell=0}^{K-1} \frac{1}{Z(\theta)} \exp\{U(Z_i = \ell, Z_{-i})\}}$$

$$PL(\theta; Z) = \prod_{i \in N} \frac{\exp\{\eta_i(z_i)\}}{\sum_{\ell=0}^{K-1} \exp\{\eta_i(\ell)\}}$$

- **The Target Pseudoposterior:** This substitution formally defines our target distribution as the **Pseudoposterior**:

$$\pi_{\text{pseudo}}(\theta \mid Z) \propto PL(\theta; Z)\pi(\theta)$$

# The Metropolis-Hastings Step via Pseudoposterior

- **Acceptance Probability:** For a symmetric proposal distribution  $D(\theta^* | \theta)$ , a proposed parameter vector  $\theta^*$  is accepted with probability  $\min(1, A)$ , where:

$$A = \frac{\pi_{\text{pseudo}}(\theta^* | Z)}{\pi_{\text{pseudo}}(\theta | Z)} = \frac{PL(\theta^*; Z)\pi(\theta^*)}{PL(\theta; Z)\pi(\theta)}$$

- In log-scale, the proposal is accepted if:

$$\log(U) < \log PL(\theta^*; Z) + \log \pi(\theta^*) - \log PL(\theta; Z) - \log \pi(\theta)$$

# The Metropolis-within-Gibbs Algorithm

- ① **Initialization:** Noise parameters  $(\mu_k, \sigma^2)$  are calculated via K-means over the observed intensity  $Y_i$  and they are used among the spacial parameters prioris to generate the initial  $Z^{(0)}$  hidden field.
- ① **Step A (Metropolis-Hastings - Field):** Propose  $(\alpha, \beta, \delta)$  via a random walk, evaluated over the **Pseudolikelihood (PL)** (but can also be initialized by user-chosen prioris).
- ② **Step B (Metropolis-Hastings - Seeds):** Propose discrete 1-pixel steps for the coordinates  $(x, y)$  of the seeds.
- ③ **Step C (Gibbs - Latent Field):** Sample  $Z_i^{(t)}$  conditioned on its neighbors and the observed intensity  $Y_i$ .
- ④ **Step D (Gibbs - Noise):** Sample new  $(\mu_k, \sigma^2)$  directly from closed-form analytical distributions.

# Full Conditionals and Conjugacy

- **Updated Posterior Mean ( $\mu_k$ ):** The full conditional under conjugate priors is a Normal:

$$m_{\text{post}} = \frac{\frac{m_k}{\tau_k^2} + \frac{\sum_{i:Z_i=k} y_i}{\sigma^2}}{\frac{1}{\tau_k^2} + \frac{n_k}{\sigma^2}}$$

$$\tau_{\text{post}} = \left( \frac{1}{\tau_k^2} + \frac{n_k}{\sigma^2} \right)^{-1}$$

- **Updated Posterior Variance ( $\sigma^2$ ):** Follows an Inverse-Gamma:

$$a_{\text{post}} = a_{\sigma} + \frac{n_k}{2}$$

$$b_{\text{post}} = b_{\sigma} + \frac{1}{2} \sum_{i:Z_i=k} (y_i - \mu_k)^2$$

# Code Engineering & Stability Challenges

- **Rcpp for Real-World Grids:** Recreated the local neighborhood and distance calculations in **C++ via Rcpp** to make iterations highly efficient.

## 5. Application

# Image Segmentation Application

- **The test:** In order to evaluate our model in real world data, a near-infrared image by Aslahishahri et al. (2021) was taken for testing.
- **Prioris:**

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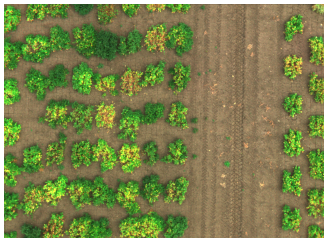
|                       |                          |
|-----------------------|--------------------------|
| ▪ $\alpha$ shape = 2  | ▪ $\mu$ mean (m) = 0     |
| ▪ $\alpha$ rate = 1   | ▪ $\mu$ sd (tau) = 0.001 |
| ▪ $\beta$ shape = 2   | ▪ $\sigma$ a = 10        |
| ▪ $\beta$ rate = 1    | ▪ $\sigma$ b = 10        |
| ▪ $\delta$ shape = 10 | ▪ $\delta$ rate = 0.5    |

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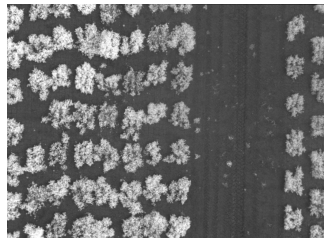
- **Initialization:**

- $\alpha = 2$
- $\beta = 2$
- $\delta_{1,2,3} = 30$

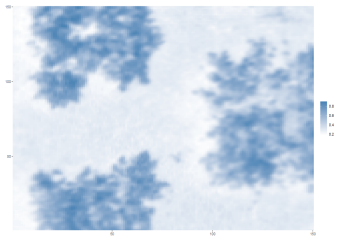
# Image Segmentation



(a) Real colored image



(b) Near-infrared image



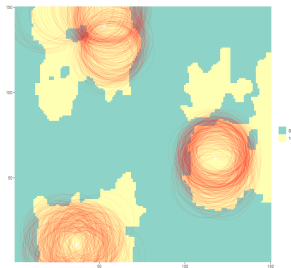
(c) Image frame



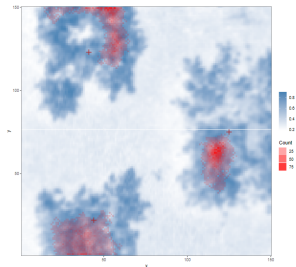
(d) Z initialization



# Seed Locations



(a) Influence area estimated locations



(b) Seeds random walk heatmap

# Image Segmentation Application Results

- **Good initialization:** Testing prioris may lead to a good first  $Z$ .
- **Fast Convergence:** After step 30001, parameters did not change drastically. The entire pipeline took  $\sim 7$  minutes for 10000 steps.
- **Accurate seeds:** Seed plots showed that seeds were well estimated as well as their influence area, but could get confused with background-within cases.

# Conclusion

- **Key Conclusions:** The MWG pipeline makes inference in spacially complex HMRFs viable and robust.
- **Efficiency:** Estimation via Pseudoposterior balances computational cost and statistical precision.

# Acknowledgements & Contact

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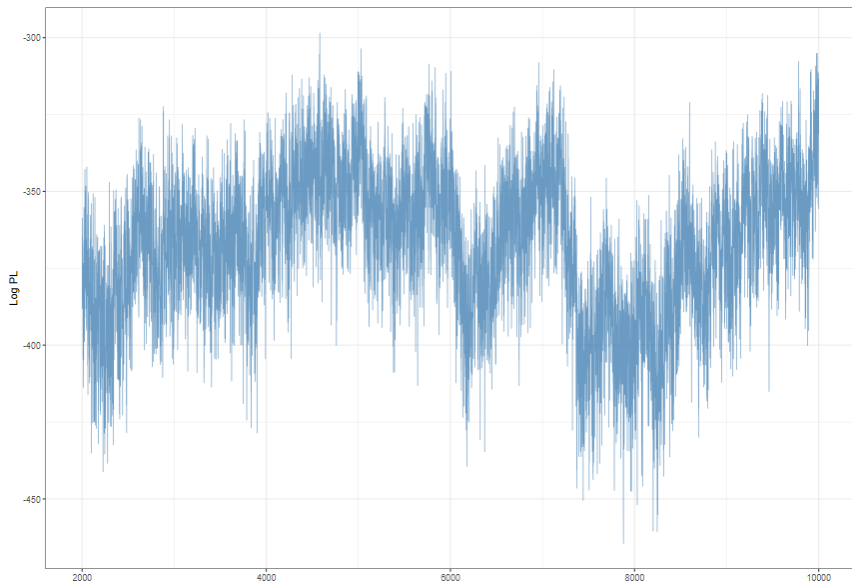
## Questions & Discussion

Thank you for your attention! Feel free to reach out for questions.

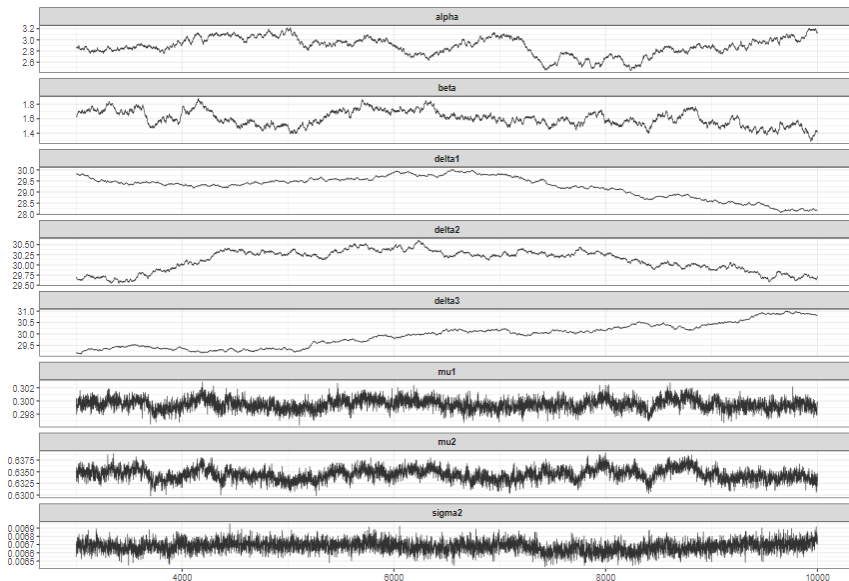
# References

Aslahishahri, Masoomah, Kevin G Stanley, Hema Duddu, Steve Shirtliffe, Sally Vail, Kirstin Bett, Curtis Pozniak, and Ian Stavness. 2021. "From RGB to NIR: Predicting of Near Infrared Reflectance from Visible Spectrum Aerial Images of Crops." In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 1312–22.

# Parameters Convergence



# Parameters Convergence



# Parameters Convergence

